

Medical Insurance Charges Prediction using Supervised Machine Learning

Project Report

Submitted By:

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1. Title

Medical Insurance Charges Prediction using Supervised Machine Learning

2. Problem Statement

Health insurance companies need to estimate the expected medical expenses of customers before issuing insurance policies. These expenses depend on several factors such as age, gender, Body Mass Index (BMI), smoking habits, number of children, and region of residence. Predicting medical insurance charges manually can be difficult because these factors influence healthcare costs in different ways.

The objective of this project is to develop a supervised machine learning model that predicts medical insurance charges using demographic and lifestyle information. The project also compares different regression algorithms to determine the most suitable model for this prediction task.

3. Dataset Description

The dataset used in this project is the **Medical Cost Personal Dataset**, which contains information about individuals and their corresponding medical insurance charges. It consists of **1,338 records** and **7 attributes**.

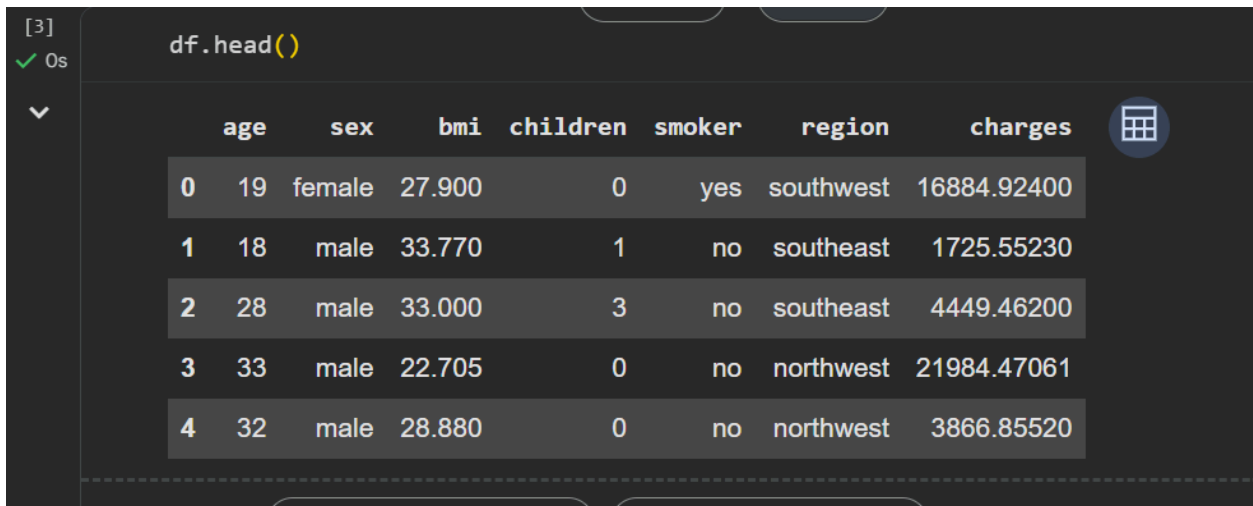
The dataset includes the following features:

- Age
- Gender
- Body Mass Index (BMI)
- Number of Children
- Smoking Status
- Residential Region
- Medical Insurance Charges (Target Variable)

The dataset was obtained from Kaggle.

Dataset Link:

[Medical Cost Personal Datasets](#)



[3] ✓ Os

df.head()

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

4. Data Preprocessing Steps

Before training the machine learning models, several preprocessing steps were performed. The dataset was loaded using Pandas.

Dataset information and data types were examined.

Missing values were checked.

Duplicate records were identified and removed.

Categorical variables (sex, smoker, and region) were converted into numerical values using Label Encoding.

The dataset was divided into training and testing sets.

Feature scaling was performed using StandardScaler.

```
[5] df.info()
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1338 non-null   int64
1   sex         1338 non-null   object
2   bmi         1338 non-null   float64
3   children    1338 non-null   int64
4   smoker      1338 non-null   object
5   region      1338 non-null   object
6   charges     1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```
[7] df.isnull().sum()
...
age      0
sex      0
bmi      0
children 0
smoker   0
region   0
charges  0
dtype: int64
```

```
[8] df.duplicated().sum()
... np.int64(1)
```

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was carried out to understand the dataset and identify patterns between different variables.

The following visualizations were generated:

- Distribution of Insurance Charges
- Age Distribution
- BMI Distribution
- Charges by Smoking Status
- Correlation Heatmap

Observations

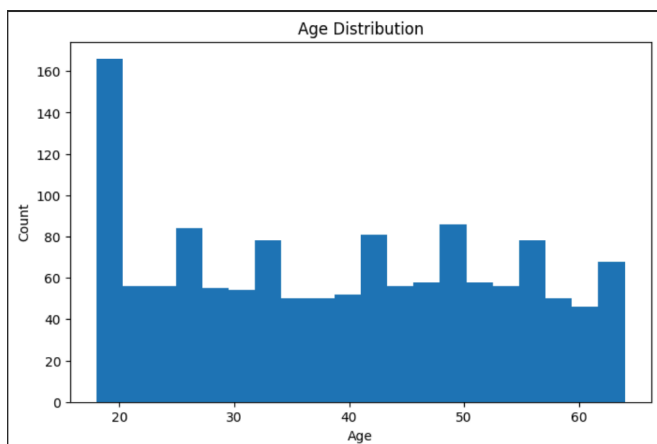
- Smokers generally have significantly higher medical insurance charges than non-smokers.
- Medical expenses tend to increase with age.
- Individuals with higher BMI values usually have higher insurance charges.
- Smoking status was found to be one of the strongest factors affecting insurance costs.

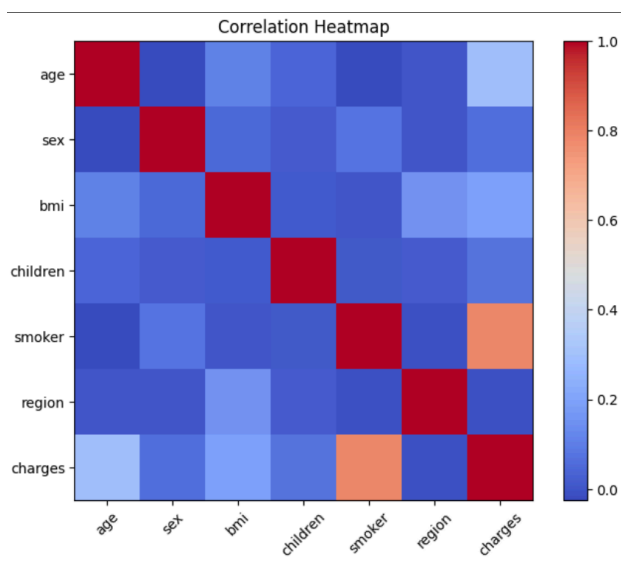
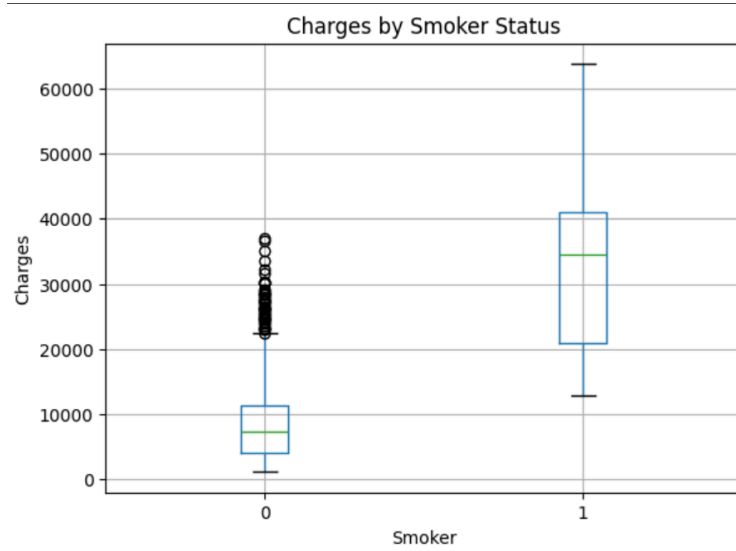
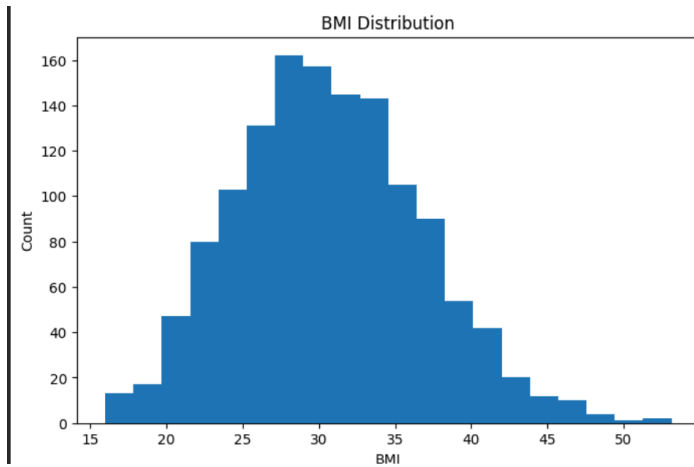
```
[11]
✓ Os
le = LabelEncoder()

df["sex"] = le.fit_transform(df["sex"])
df["smoker"] = le.fit_transform(df["smoker"])
df["region"] = le.fit_transform(df["region"])

df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	0	27.900	0	1	3	16884.92400
1	18	1	33.770	1	0	2	1725.55230
2	28	1	33.000	3	0	2	4449.46200
3	33	1	22.705	0	0	1	21984.47061
4	32	1	28.880	0	0	1	3866.85520





6. Model Used

This is a **Supervised Machine Learning Regression** problem because the target variable (insurance charges) is continuous.

Three regression algorithms were implemented and compared:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Among these models, **Random Forest Regressor** achieved the best prediction performance.

7. Training Process

The complete machine learning workflow followed in this project is shown below:

Dataset
↓
Data Preprocessing
↓
Feature Encoding
↓
Feature Scaling
↓
Train-Test Split
↓
Model Training
↓
Prediction
↓
Model Evaluation

The dataset was divided into 80% training data and 20% testing data. Each regression model was trained on the training dataset and evaluated using the testing dataset.

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

8. Evaluation Metrics and Results

The following regression metrics were used:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

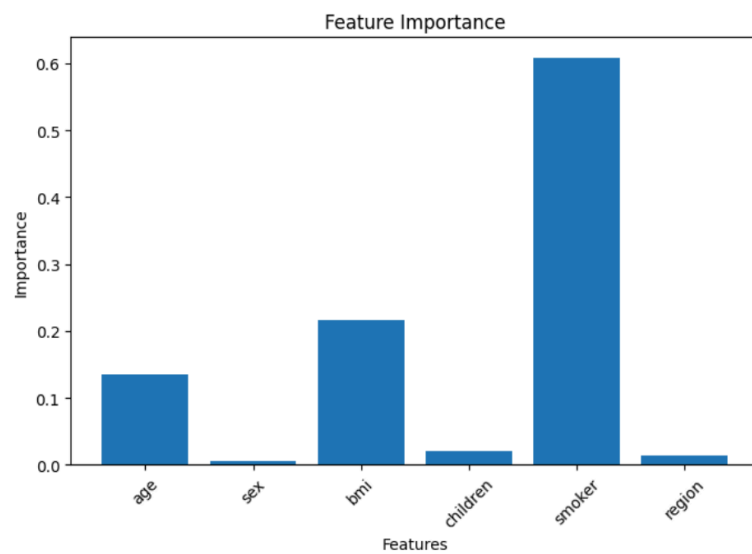
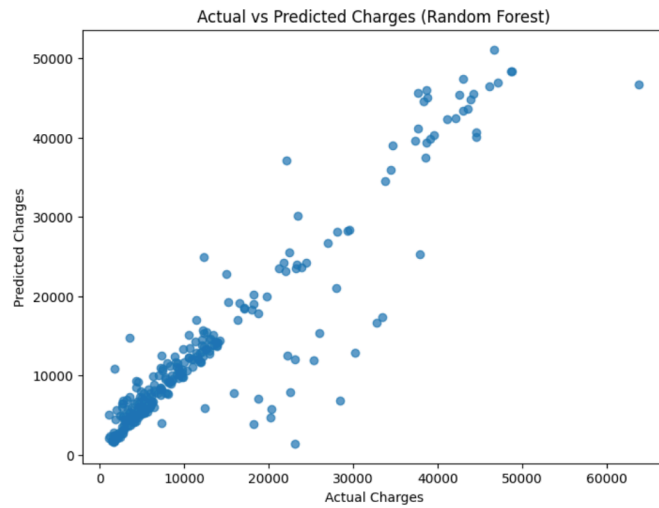
Linear Regression Results

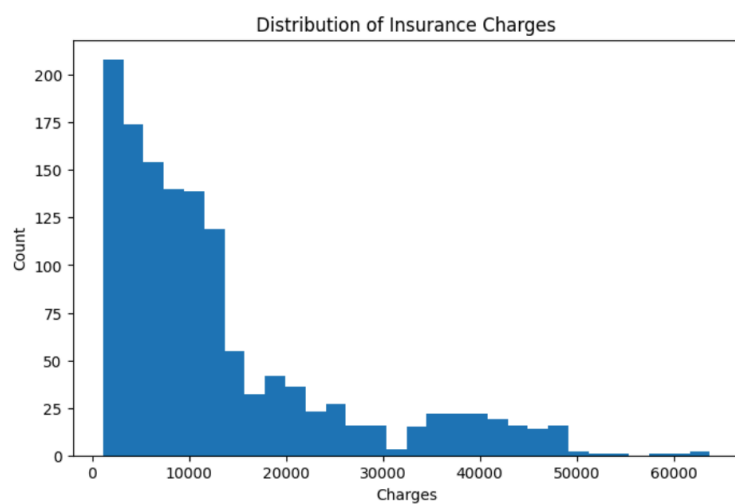
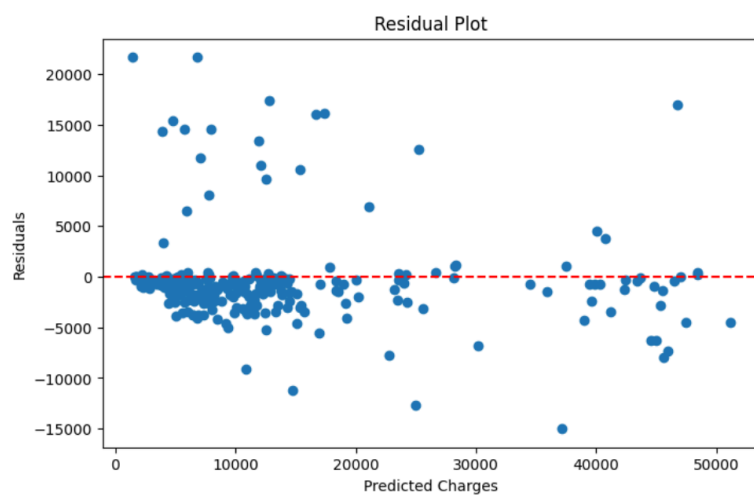
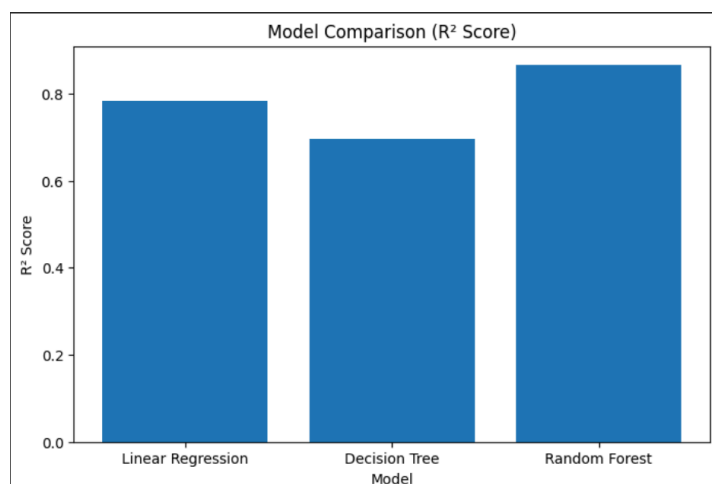
Metric	Value
MAE	4186.51
MSE	33,635,210.3
RMSE	5799.59
R^2 Score	0.7833

The Decision Tree and Random Forest models were also trained and evaluated. Random Forest produced the highest prediction accuracy and was selected as the best-performing model.

```
[22]
✓ Os
print("Linear Regression Results")
print("MAE :", mean_absolute_error(y_test, lr_pred))
print("MSE :", mean_squared_error(y_test, lr_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, lr_pred)))
print("R2 Score:", r2_score(y_test, lr_pred))
```

Linear Regression Results
MAE : 4186.508898366436
MSE : 33635210.43117845
RMSE: 5799.5870914383595
R2 Score: 0.7833463107364536





[35] ✓ 0s

results

	Model	MAE	RMSE	R2 Score
0	Linear Regression	4186.508898	5799.587091	0.783346
1	Decision Tree	3070.083148	6881.111205	0.695008
2	Random Forest	2518.475635	4571.501475	0.865386

```

new_customer = pd.DataFrame({
    "age": [25],
    "sex": [1],
    "bmi": [28.5],
    "children": [0],
    "smoker": [0],
    "region": [2]
})

new_customer_scaled = scaler.transform(new_customer)

predicted_charge = rf.predict(new_customer_scaled)

print("Predicted Insurance Charge:", predicted_charge[0])

... Predicted Insurance Charge: 2530.180055999999

```

9. Conclusion

This project successfully developed a supervised machine learning model capable of predicting medical insurance charges using demographic and lifestyle information.

Among the three regression models implemented, the Random Forest Regressor provided the best predictive performance. The analysis showed that smoking status, age, and BMI have the greatest influence on insurance charges.

This project demonstrates how supervised machine learning can assist insurance organizations in estimating expected healthcare expenses and supporting data-driven decision-making.

Future improvements may include using larger datasets, incorporating additional health-related features, and deploying the trained model as a web application for real-time prediction.

10. References

Kaggle – Medical Cost Personal Dataset
<https://www.kaggle.com/datasets/mirichoi0218/insurance>
